

MODULE II

**INTRODUCTION TO COMPUTER FORENSICS
INVESTIGATION AND ELECTRONIC
EVIDENCE**

COMPUTER FORENSICS

- ☐ Computer forensics involves the **preservation, identification, extraction, documentation, and interpretation of computer media for evidentiary and/or root cause analysis.**
- ☐ Evidence might be required for a wide range of computer crimes and misuses.
- ☐ It involves:
 1. **Discovering data on computer system**
 2. **Recovering deleted, encrypted, or damaged file information**
 3. **Monitoring live activity**
 4. **Detecting violations of corporate policy**
- ☐ Information collected assists in arrests, prosecution, termination of employment, and preventing future illegal activity.

EXAMPLES

- ❑ Recovering thousands of **deleted emails**.
- ❑ Performing **investigation after multiple users** had taken over the system.
- ❑ **Performing termination investigation post employment.**
- ❑ Recovering **evidence post formatting hard drive.**

WHO USES COMPUTER FORENSICS

Criminal Prosecutors :Rely on evidence obtained from a computer to prosecute suspects and use as evidence.

Civil Litigations :Personal and business data discovered on a computer can be used in fraud, divorce, harassment, or discrimination cases.

Insurance Companies :Evidence discovered on computer can be used to mollify costs, worker's compensation, arson, etc.

Private Corporations :Obtained evidence from employee computers can be used as evidence in harassment, fraud, and embezzlement cases.

Law Enforcement Officials :Rely on computer forensics to backup search warrants and post seizure handling.

Individual/Private Citizens :Obtain the services of professional computer forensic specialists to support claims of harassment, abuse, or wrongful termination from employment.

TYPES OF DIGITAL FORENSICS

Disk Forensics: It deals with extracting data from storage media by searching active, modified, or deleted files.

Network Forensics: It is a sub branch of digital forensics. It is related to monitoring and analysis of computer network traffic to collect important information and legal evidence.

Wireless Forensics: It is a division of network forensics. The main aim of wireless forensics is to offers the tools need to collect and analyze the data from wireless network traffic.

Malware Forensics: This branch deals with the identification of malicious code, to study their payload, viruses, worms, etc.

Email Forensics: Deals with recovery and analysis of emails, including deleted emails, calendars, and contacts.

Memory Forensics: It deals with collecting data from system memory (system registers, cache, RAM) in raw form and then carving the data from Raw dump.

Mobile Phone Forensics: It mainly deals with the examination and analysis of mobile devices. It helps to retrieve phone and SIM contacts, call logs, incoming, and outgoing SMS/MMS, Audio, videos, etc.

Database Forensics: It is a branch of digital forensics relating to the study and examination of databases and their related metadata.

IoT Forensics: IoT forensics is the practice of analyzing IoT devices to investigate crimes. Fitness trackers, smart appliances, connected vehicles, would form part of IoT Forensics.

Cloud Forensics: Cloud forensic is the amalgamation of all the different forensics(i.e digital forensics, network forensics, hardware forensics etc.

DIGITAL EVIDENCE

- ☐ Any information being subject to human intervention or not, that can be extracted from a computer.
- ☐ Must be in human-readable format or capable of being interpreted by a person with expertise in the subject.

REASONS FOR EVIDENCE GATHERING

Evidence collected by Federal, State and local authorities for crimes relating to:

- Theft of trade secrets
- Fraud
- Extortion
- Industrial espionage
- Position of pornography
- SPAM investigations
- Virus/Trojan distribution

- Homicide investigations
- Intellectual property breaches
- Unauthorized use of personal information
- Unauthorized activity
- Tracking internet browsing habits
- Reconstructing Events
- Inferring intentions
- Selling company bandwidth
- Wrongful dismissal claims
- Sexual harassment
- Software Piracy

LOCARD'S PRINCIPLE OF EXCHANGE

- Locard's exchange principle is an important part of forensic science investigation.
- **It states that any criminal leaves behind a trace when committing a violent crime.**
- It is the investigator's duty to find this trace evidence and reconstruct the events of the crime.
- **In other words, the perpetrator of a crime will bring something into the crime scene and leave with something from it, and that both can be used as forensic evidence.**

STEPS OF COMPUTER FORENSICS

- Digital forensics entails the following steps:

Identification

- Identify the purpose of investigation
- Identify the resources required

Preservation

- Data is isolate, secure and preserve

Analysis

- Identify tool and techniques to use
- Process data
- Interpret analysis results

Documentation

- Documentation of the crime scene along with photographing, sketching, and crime-scene mapping

Presentation

- Process of summarization and explanation of conclusions is done with the help to gather facts.

PROCESS OF DIGITAL FORENSICS

Identification: It is the **first step in the forensic process**. The identification process mainly includes things like **what evidence is present, where it is stored, and lastly, how it is stored (in which format)**. Electronic storage media can be personal computers, mobile phones, PDAs, etc.

Preservation: In this phase, data is **isolated, secured, and preserved**. It includes preventing people from using the digital device so that digital evidence is not tampered with.

Analysis: In this step, **investigation agents reconstruct fragments of data and draw conclusions based on evidence found**.

Documentation

- In this process, a record of all the visible data must be created.
- It helps in recreating the crime scene and reviewing it.
- It Involves **proper documentation of the crime scene along with photographing, sketching, and crime-scene mapping.**

Presentation

- In this last step, the **process of summarization and explanation of conclusions is done.**

DIGITAL CRIME SCENE INVESTIGATION PROCESS

- A **digital investigation** is a **process** where **hypotheses that answer questions about digital events** is developed and tested.
- A digital forensic investigation is a **process that uses science and technology to analyze digital objects** and that develops and tests theories.
- It **should be admissible in a court of law**, to answer questions about **events that occurred**.
- In other words, **a digital forensic investigation is a more restricted form of digital investigation**.

Digital crime scene includes digital environment created by hardware and software.

Digital Crime Investigation process has three major phases:

- 1. System Preservation Phase**
- 2. Evidence Searching Phase**
- 3. Event Reconstruction Phase**

TYPES OF ANALYSIS:

Live Analysis

Dead Analysis

INITIATING A INVESTIGATION

1. **Do not begin by exploring files on system randomly.**
2. Establish evidence custodian - start a detailed journal with the date and time and date/information discovered.
3. If possible, **designate suspected equipment as “off- limits” to normal activity.** This includes back-ups and configuration changes.
4. Collect email, DNS, and other network service logs.
5. Capture exhaustive external TCP and UDP port scans of the host.
6. Contact security personnel [CERT], management, State and local enforcement, as well as affected sites or persons.

PRESERVATION PHASE

- Involves **preservation of state of digital crime scene.**
- Action taken is based on legal or operational requirements.
- **Unplug the system to create a image.**
- **Purpose** – To reduce the amount of evidence that could be overwritten.
- **Write blocker can be used to prevent overwriting of data.**
- Creating a cryptographic hash of the data like MD5, SHA1, etc.

HANDLING EVIDENCE

❖ **Admissibility of Evidence**

- Legal rules which determine **whether potential evidence can be considered by a court.**
- **Must be obtained in a manner which ensures the authenticity and validity and that no tampering had taken place.**

- ❖ Preventing viruses from being introduced to a computer during the analysis process.
- ❖ Extracted/relevant evidence is properly handled and protected from mechanical or electromagnetic damage.

- ❖ No possible evidence is damaged, destroyed, or otherwise compromised by the procedures used to search the computer.
- ❖ Establishing and maintaining a continuing chain of custody.
- ❖ Limiting the amount of time business operations are affected.
- ❖ **Not divulging and respecting any ethically [and legally] client-attorney information that is inadvertently acquired during a forensic exploration.**

HANDLING INFORMATION

- Information and data being sought after and collected in the investigation must be properly handled.

✓ Volatile Information

- **Network Information**
 - Communication between system and the network.
- **Active Processes**
 - Programs currently active on the system.
- **Logged-on Users**
 - Users/employees currently using system.
- **Open Files**
 - Libraries in use; hidden files; Trojans (rootkit) loaded in system.

✓ **Non-Volatile Information**

- This includes information, **configuration settings, system files and registry settings that are available after reboot.**
- Accessed through drive mappings from system.
- This information should investigated and reviewed from a backup copy.

EVIDENCE GATHERING PHASE

- After data is preserved, **evidence needs to be searched.**
- Depending the evidence type **various locations are searched.**
- The hypothesis **created based on the case details need to be refuted or supported to ensure appropriateness of evidence collected.**
- General characteristics of the object being searched needs to be defined and then searched in the data gathered.
- Evidence can be **searched based on name, pattern, comparing hash, keyword based or searching for IP address, specific port/source address.**

EVENT RECONSTRUCTION PHASE

- Based on the evidence found reconstruction of the digital event is done in this phase.
- Once digital event reconstruction is done, **it can be compared with physical events.**
- **Event reconstruction requires adequate knowledge of OS and the installed applications.**

GENERAL GUIDELINES

PICL should be followed in all investigations to ensure no evidence is left out

- **P – Preservation**
- **I – Isolation**
- **C - Correlation**
- **L – Logging**

Preservation

- Original data should be kept in safe custody and investigation should be on the copy of the original data.
- Write blocker should be used.
- Calculate hash of the original evidence.
- Live analysis should be done carefully to prevent overwriting of existing data.

ISOLATION

- The analysis environment **should be isolated from all possible threats.**
- The analysis should be done in a **virtual environment to prevent any data loss.**
- **Connection to the outside world should be avoided to prevent any tampering of evidence.**
- Isolation is implemented using an analysis network that has limited connectivity.
- Implementing isolation during a live analysis is difficult and hence should be done with caution.

CORRELATION

- Data should be correlated with other independent sources to prevent any forgery/planting of evidence.
- **Timestamps can be easily manipulated and preventing correlation in such instances.**
- **File activity timeline should be correlated with log entries, network traffic or other events.**

LOGGING

- All the actions being taken should be logged and documented.
- This will prevent missing important actions and activities.
- **Data changes during live analysis should be well documented.**

STANDARD OPERATING PROCEDURES

- **First responders may follow the steps listed below to guide their handling of digital evidence at an electronic crime scene:**
 - Recognize, identify, seize, and secure all digital evidence at the scene.
 - Document the entire scene and the specific location of the evidence found.
 - Collect, label, and preserve the digital evidence.
 - Package and transport digital evidence in a secure manner.

Before collecting evidence at a crime scene, first responders should ensure that:

- Legal authority exists to seize evidence.
- The scene has been secured and documented.
- Appropriate personal protective equipment is used.

First responders without the proper training and skills should not attempt to explore the contents of or to recover information from a computer or other electronic device other than to record what is visible on the display screen.

- **For proper evidence preservation, follow these procedures in order:**
 - Photograph the computer and scene.
 - If the computer is off , do not turn it on.
 - If the computer is on, photograph the screen.
 - **Collect live data - start with RAM image** and then collect other live data "as required" such as network connection state, logged on users, currently executing processes etc.
 - **Unplug the power cord from the back of the tower** - If the computer is a laptop and does not shut down when the cord is removed then remove the battery.

- **Document all device model numbers and serial numbers.**
- Disconnect all cords and devices.
- Image hard drives using a write blocker.
- Package all components(using anti-static evidence bags).
- **Seize all additional storage media** (create respective images and place original devices in anti-static evidence bags).
- **Keep all media away from magnets, radio transmitters and other potentially damaging elements.**
- Collect instruction manuals, documentation and note.
- Document all steps used in the seizure and **maintain proper Chain of Custody.**

CHAIN OF CUSTODY

- **The movement and location of physical evidence from the time it is obtained until the time it is presented in court.**
- **As is the case with all evidence, it's important to maintain a chain of custody for computer evidence.**
- **The term "chain of custody" refers to documentation that identifies all changes in the control, handling, possession, ownership, or custody of a piece of evidence (physical or electronic).**
- **It is required to trace the route that evidence takes from the moment it is collected until the time it is presented in Court of Law.**

- Chain of Custody refers to the **logical sequence that records the sequence of custody, control, transfer, analysis and disposition of physical or electronic evidence in legal cases.**
- Each step in the chain is essential and if there is a break, the evidence may be rendered inadmissible.

EVIDENCE

Agency: _____

Agent: : _____

Case #: _____ **Item #:** _____

Description: _____

Location: _____

Remarks: _____

CHAIN OF CUSTODY

From	To	Date

EVIDENCE CHAIN OF CUSTODY TRACKING FORM

Case Number: _____ Offense: _____

Submitting Officer: (Name/ID#) _____

Victim: _____

Suspect: _____

Date/Time Seized: _____ Location of Seizure: _____

Description of Evidence		
Item #	Quantity	Description of Item (Model, Serial #, Condition, Marks, Scratches)

Chain of Custody				
Item #	Date/Time	Released by (Signature & ID#)	Received by (Signature & ID#)	Comments/Location

IMPORTANCE OF CHAIN OF CUSTODY

- Maintaining the chain of custody is critical in forensic practice **to avert tampering.**
- The goal is to establish that the evidence is related to the alleged crime, was collected from the scene, and was in **its original/unaltered condition rather than having been tampered with or "deceitfully to make someone seem guilty.**
- The chain of **custody maintains the integrity of the sample.**
- The **traceability of the record of the control, transfer, and analysis of samples indicates the transparency to the procedure.**

WRITE BLOCKERS

A write blocker is any **tool that permits read only access to data storage devices without compromising the integrity of the data.**

It prevents any write access to the hard disk.

Write blockers are devices that allow acquisition of information on a drive without creating the possibility of accidentally damaging the drive contents. **They do this by allowing read commands to pass but by blocking write commands.**

As per NIST general guidelines:

- ❖ The write blocker tool shall not **allow a protected drive to be changed.**
- ❖ The write blocker tool shall not prevent **any operations to a drive.**
- ❖ The write blocker tool shall not **prevent obtaining any information from or about any drive.**

TYPES OF WRITE BLOCKERS

Write Blockers are basically of 2 types: Hardware Write Blocker and Software Write Blocker

- **Hardware write blocker**—The hardware blocker is a device that is installed that runs software internally to itself and will block the write capability of the computer to the device attached to the write blocker.
- **Software write blocker**—The software blocker is an application that is run on the operating system that implements a software control to turn off the write capability of the operating system.

